

```
In [1]: import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt
```

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In [4]: #Task 1, 2, 3 – Create Correlation Matrix along with Create Heatmap with Annotations and Diverging color palette  
  
df = pd.read_csv("food_delivery_orders (1).csv")
```

```
In [5]: df
```

Out[5]:

	delivery_time	order_value	tip_amount	rating	delivery_distance
0	20	250	20	4.9	2.0
1	22	300	25	4.8	2.5
2	24	280	18	4.8	2.2
3	25	350	30	4.7	3.0
4	27	400	35	4.7	3.2
5	28	380	28	4.6	3.5
6	30	450	40	4.6	3.8
7	31	500	45	4.5	4.0
8	33	420	32	4.5	4.2
9	35	550	50	4.4	4.5
10	36	600	55	4.3	4.8
11	38	520	42	4.3	5.0
12	40	650	60	4.2	5.2
13	42	700	65	4.1	5.5
14	44	620	58	4.1	5.8
15	45	750	70	4.0	6.0
16	47	800	75	3.9	6.3
17	49	720	68	3.8	6.5
18	52	900	90	3.7	7.0
19	55	850	80	3.5	7.5

In [7]: correlation = df.corr(numeric_only=True)

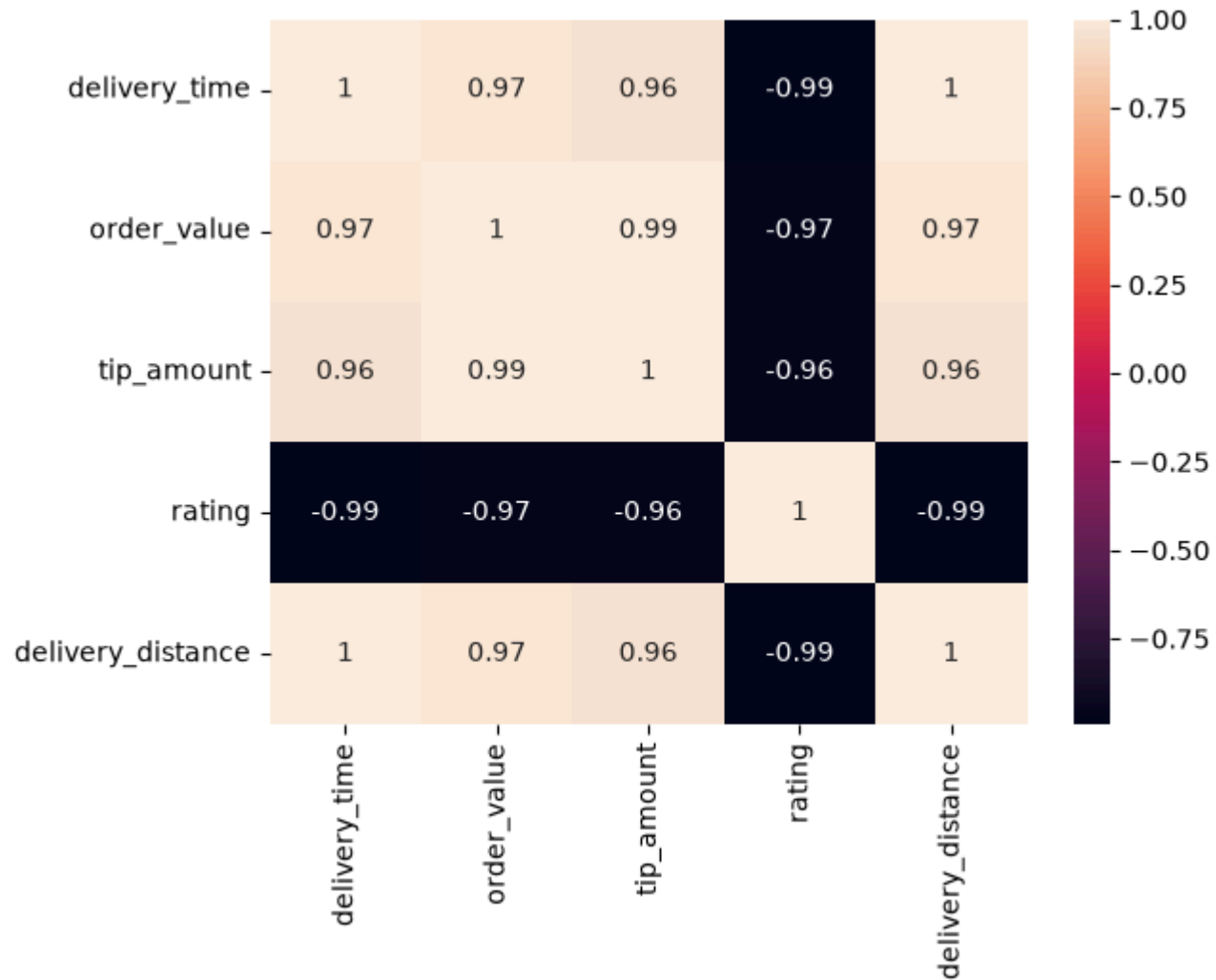
```
print(correlation)
```

	delivery_time	order_value	tip_amount	rating	\
delivery_time	1.000000	0.972358	0.955866	-0.993680	
order_value	0.972358	1.000000	0.994968	-0.971217	
tip_amount	0.955866	0.994968	1.000000	-0.957457	
rating	-0.993680	-0.971217	-0.957457	1.000000	
delivery_distance	0.995231	0.974514	0.958831	-0.989294	

	delivery_distance
delivery_time	0.995231
order_value	0.974514
tip_amount	0.958831
rating	-0.989294
delivery_distance	1.000000

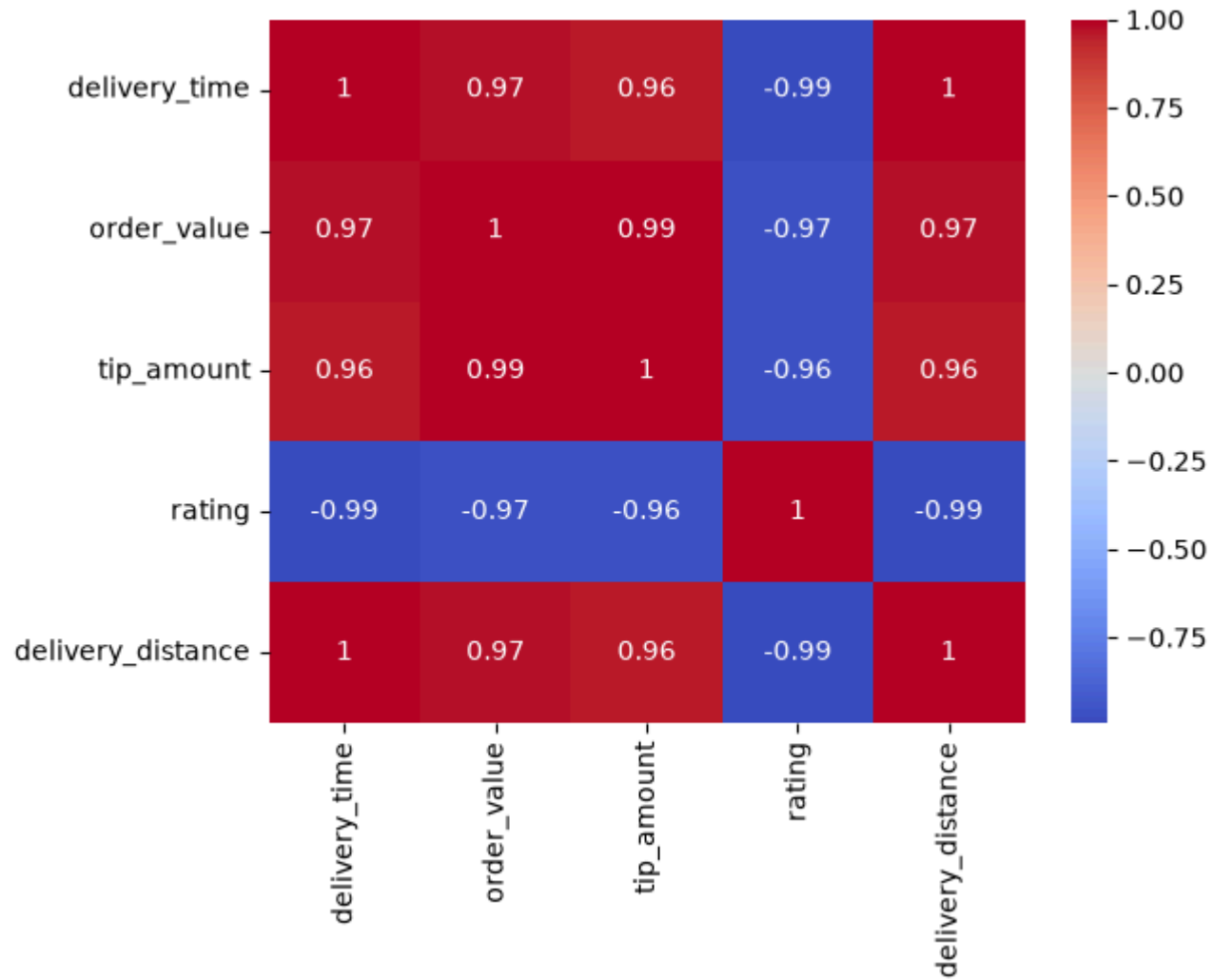
```
In [12]: sns.heatmap(correlation, annot=True)
```

```
Out[12]: <Axes: >
```



```
In [13]: sns.heatmap(correlation, annot=True, cmap="coolwarm")
```

```
Out[13]: <Axes: >
```



```
In [14]: print(correlation)
```

	delivery_time	order_value	tip_amount	rating	\
delivery_time	1.000000	0.972358	0.955866	-0.993680	
order_value	0.972358	1.000000	0.994968	-0.971217	
tip_amount	0.955866	0.994968	1.000000	-0.957457	
rating	-0.993680	-0.971217	-0.957457	1.000000	
delivery_distance	0.995231	0.974514	0.958831	-0.989294	

	delivery_distance
delivery_time	0.995231
order_value	0.974514
tip_amount	0.958831
rating	-0.989294
delivery_distance	1.000000

#Conclusion- Delivery time and rating have the strongest negative correlation. This means that as delivery time increases, the rating generally decreases in this sample dataset.

In [15]: *#Task 4 – Flipkart Mobile Phones*

```
df = pd.read_csv("flipkart_mobile_phones.csv")
```

In [16]: df

Out[16]:

	model	price	rating	ram	battery_capacity	storage
0	Phone A	12000	4.1	4	5000	64
1	Phone B	15000	4.2	4	5000	128
2	Phone C	18000	4.0	6	4500	128
3	Phone D	22000	4.3	6	5000	128
4	Phone E	25000	4.2	8	5000	256
5	Phone F	28000	4.4	8	4800	256
6	Phone G	32000	4.3	8	5000	256
7	Phone H	36000	4.5	12	5200	256
8	Phone I	40000	4.4	12	5000	512
9	Phone J	45000	4.6	12	5300	512
10	Phone K	50000	4.5	16	5500	512
11	Phone L	55000	4.7	16	6000	512

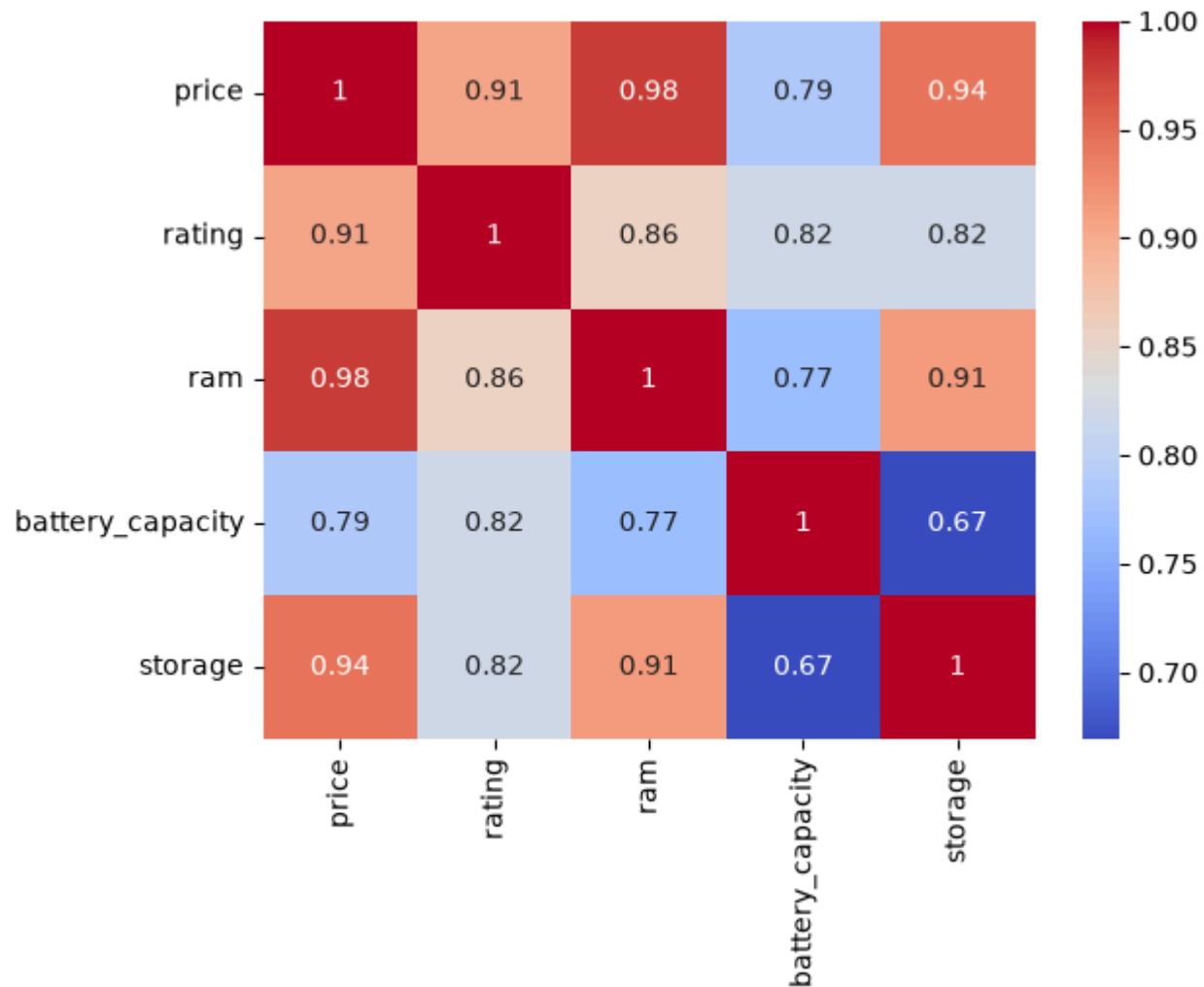
```
In [17]: correlation = df.corr(numeric_only=True)
print(correlation)
```

```

           price    rating    ram  battery_capacity  storage
price      1.000000  0.906203  0.977132      0.787612  0.943643
rating      0.906203  1.000000  0.857273      0.819801  0.818886
ram         0.977132  0.857273  1.000000      0.768933  0.913634
battery_capacity 0.787612  0.819801  0.768933      1.000000  0.668996
storage      0.943643  0.818886  0.913634      0.668996  1.000000
```

```
In [18]: sns.heatmap(correlation, annot=True, cmap="coolwarm")
```

Out[18]: <Axes: >



Conclusion The heatmap shows the correlation between price, rating, RAM, battery capacity, and storage. The numbers inside the heatmap show the strength of the relationship between the features. The coolwarm color palette makes positive and negative correlations easier to identify.